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MI&CT ESERVICE NOTE
SN07012213a

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This short-term informational note is invalid 6 months after the issue date.

Date: January 22, 2007

Subject: Slipring Brush Disconnects
Effectivity: HP-60, -80, -100 systems, RT, VCT

Symptoms: Early termination of kV, LSCOM communication issues, and missing triggers.

Resolution: See following instructions for checking disconnects, properly performing slipring cleaning, replacing signal brushes, and brushblock alignment. This information will be included in the next Service documents release.

PERSONNEL REQUIREMENTS

Required Persons	Preliminary Requirements
1	Timing Not Applicable

PRELIMINARY REQUIREMENTS

Tools and Test Equipment

Item	Qty	Effective	Part#
Flat-blade screwdriver	1	-	-
2.5mm, 4mm, 5mm, 10mm Hex Key	1	-	-
HEPA vacuum cleaner	1	-	46-297933P1

Consumables

Item	Qty	Effective	Part#	Manufacturer
Slipring eraser	1	-	5169251	Schleifring
Neoprene gloves (Large)	100	-	46-194427P347	-
Neoprene gloves (XL)	100	-	46-194427P348	-
HEPA filter (if required)	1	-	46-297948P1	-

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Replacement Parts

Item	Qty	Effective	Part#
Power brush tip set	As needed	-	2349603
Signal brush tip set	1	-	2349604

Safety

NOTICE

Follow ALL required safety and PPE procedures customary for your organization when working on this product.

Required Conditions

Condition
Only trained service personnel should service the GE CT Scanner.

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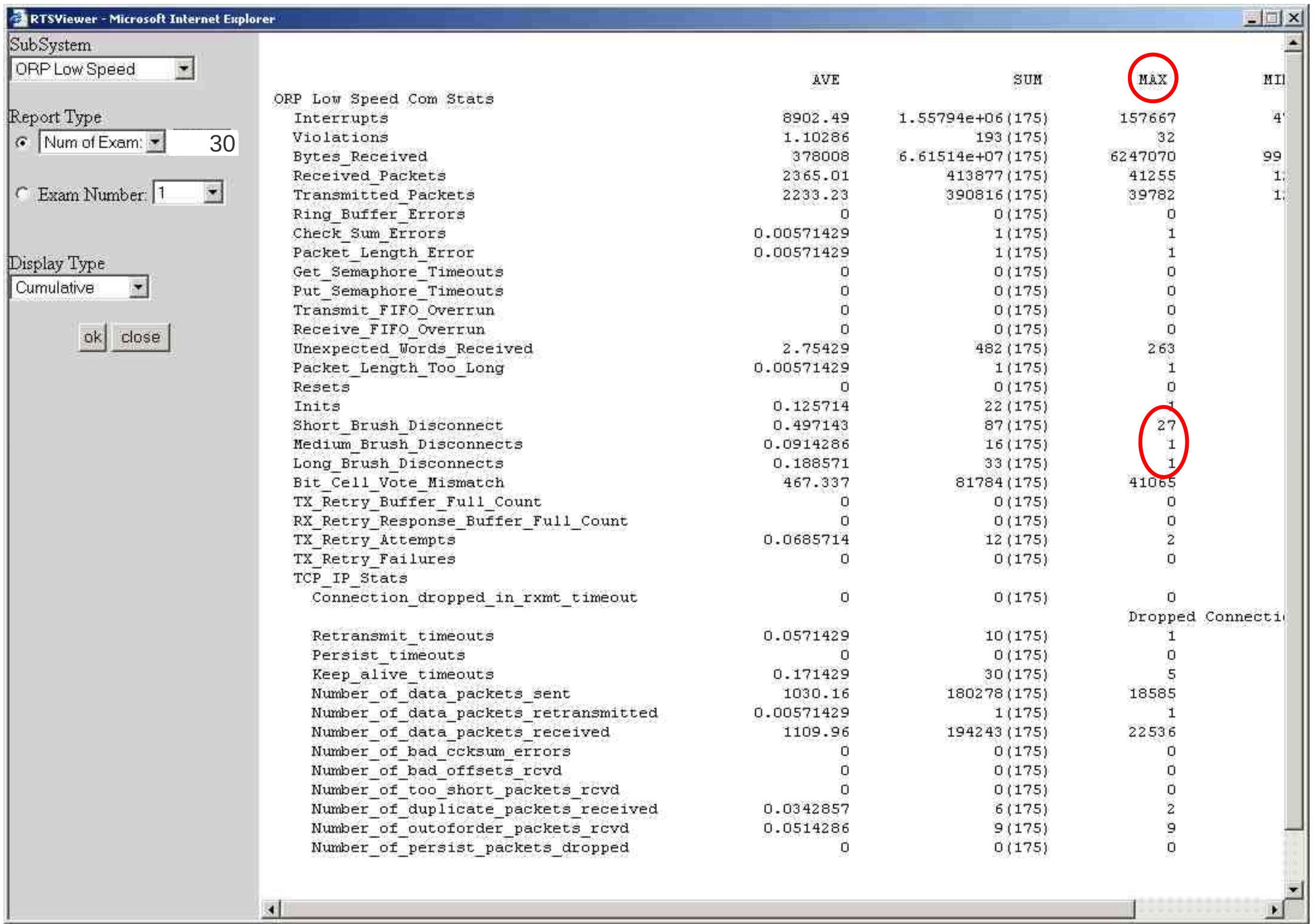
PROCEDURES

1. Check Disconnects

1-1 ORP Low Speed Disconnects

1. On the CSD, select **Diagnostics > RTS Viewer**. See Illustration 1.
   NOTE: For non-VCT systems, use *getStats*.

ILLUSTRATION 1 – ORP LOW SPEED REPORT



	AVE	SUM	MAX	MI
ORP Low Speed Com Stats				
Interrupts	8902.49	1.55794e+06(175)	157667	4
Violations	1.10286	193(175)	32	
Bytes_Received	378008	6.61514e+07(175)	6247070	99
Received_Packets	2365.01	413877(175)	41255	1
Transmitted_Packets	2233.23	390816(175)	39782	1
Ring_Buffer_Errors	0	0(175)	0	
Check_Sum_Errors	0.00571429	1(175)	1	
Packet_Length_Error	0.00571429	1(175)	1	
Get_Semaphore_Timeouts	0	0(175)	0	
Put_Semaphore_Timeouts	0	0(175)	0	
Transmit_FIFO_Overrun	0	0(175)	0	
Receive_FIFO_Overrun	0	0(175)	0	
Unexpected_Words_Received	2.75429	482(175)	263	
Packet_Length_Too_Long	0.00571429	1(175)	1	
Resets	0	0(175)	0	
Inits	0.125714	22(175)	1	
Short_Brush_Disconnect	0.497143	87(175)	27	
Medium_Brush_Disconnects	0.0914286	16(175)	1	
Long_Brush_Disconnects	0.188571	33(175)	1	
Bit_Cell_Vote_Mismatch	467.337	81784(175)	41065	
TX_Retry_Buffer_Full_Count	0	0(175)	0	
RX_Retry_Response_Buffer_Full_Count	0	0(175)	0	
TX_Retry_Attempts	0.0685714	12(175)	2	
TX_Retry_Failures	0	0(175)	0	
TCP_IP_Stats				
Connection_dropped_in_rxmt_timeout	0	0(175)	0	
			Dropped	Connecti
Retransmit_timeouts	0.0571429	10(175)	1	
Persist_timeouts	0	0(175)	0	
Keep_alive_timeouts	0.171429	30(175)	5	
Number_of_data_packets_sent	1030.16	180278(175)	18585	
Number_of_data_packets_retransmitted	0.00571429	1(175)	1	
Number_of_data_packets_received	1109.96	194243(175)	22536	
Number_of_bad_ccksum_errors	0	0(175)	0	
Number_of_bad_offsets_rcvd	0	0(175)	0	
Number_of_too_short_packets_rcvd	0	0(175)	0	
Number_of_duplicate_packets_received	0.0342857	6(175)	2	
Number_of_outoforder_packets_rcvd	0.0514286	9(175)	9	
Number_of_persist_packets_dropped	0	0(175)	0	

2. From the dropdown list in the ***SubSystem*** field select **ORP Low Speed.**
3. From the dropdown list in the ***Report Type*** field select **Num of Exams**.
4. In the field to the right of *Report Type* enter 30.
5. From the dropdown list in the ***Display Type*** field select **Cumulative**. A report appears similar to that in Illustration 1.
6. Ensure that the value in the MAX column is <= 1 disconnect per exam for short, medium, and long disconnects.
7. **If the MAX value is >1**, perform the service procedures in **Sections 4.2**–**4.5** of this document.

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Corrective Action Summary:

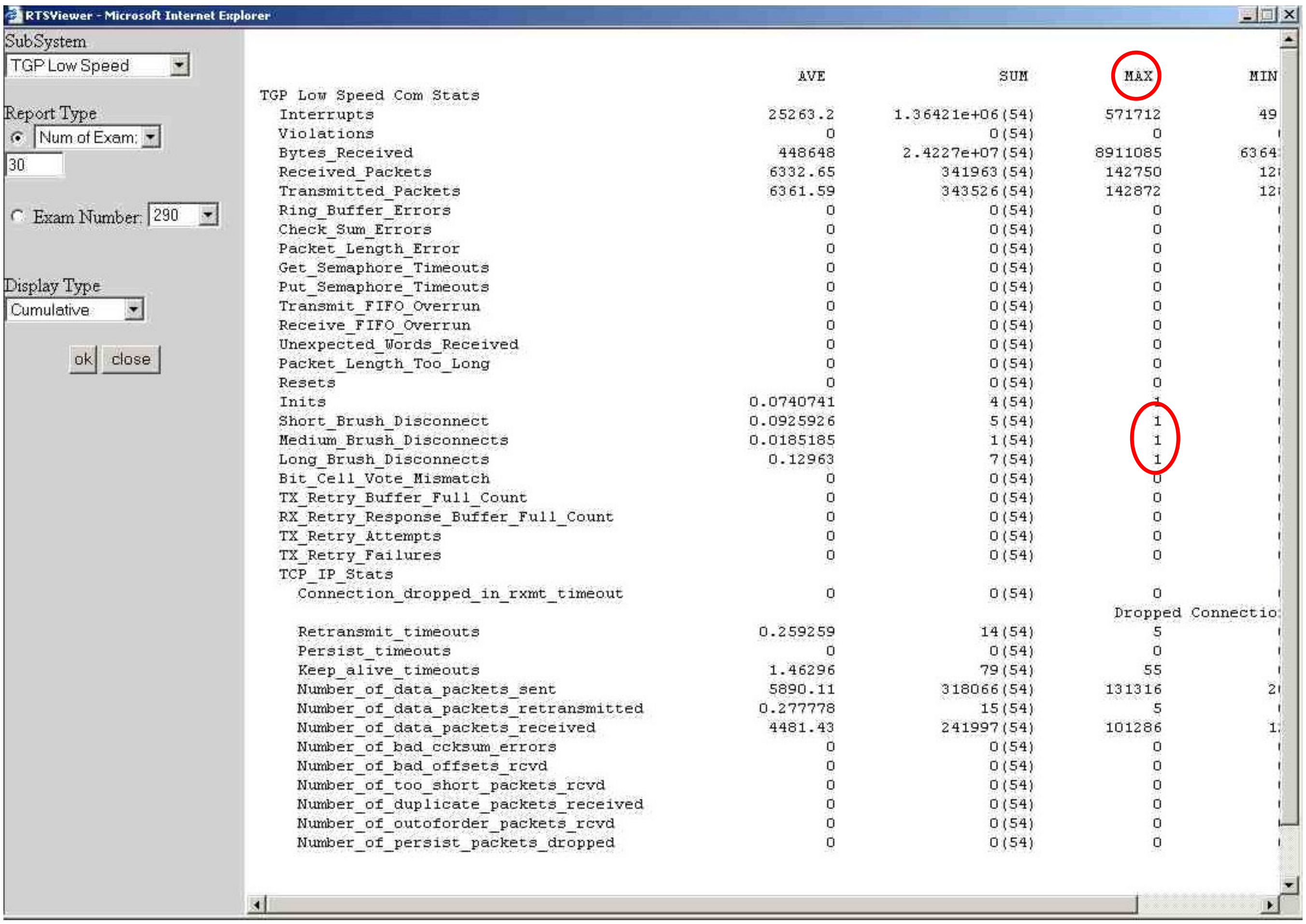
- Section 4.2 - Remove Brush Block
- Section 4.3 - Cleaning the Slipring
- Section 4.4 - Replacing the Signal Brushes
- Section 4.5 - Aligning the Brush Block

Optional procedure -> Perform the *Slipring Bypass Procedure* in Section 1-3 to test for stationary failures.

1-2 TGP Low Speed Disconnects

1. On the CSD, select **Diagnostics > RTS Viewer**. See Illustration 2.
   NOTE: For non-VCT systems, use *getStats*.

ILLUSTRATION 2 - TGP LOW SPEED REPORT



8. From the dropdown list in the ***SubSystem*** field select **TGP Low Speed.**
9. From the dropdown list in the ***Report Type*** field select **Num of Exams**.
10. In the field to the right of *Report Type* enter 30.
11. From the dropdown list in the ***Display Type*** field select **Cumulative**. A report appears similar to that in Illustration 2.

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12. Ensure that the value in the MAX column is <= 1 disconnect per exam for short, medium, and long disconnects.
13. **If the MAX value is >1**, perform the service procedures in **Sections 4.2**–**4.5** of this document.

<u>Corrective Action Summary:</u>

- Section 4.2 - Remove Brush Block
- Section 4.3 - Cleaning the Slipring
- Section 4.4 - Replacing the Signal Brushes
- Section 4.5 - Aligning the Brush Block

Optional procedure -> Perform the *Slipring Bypass Procedure* in Section 1-3 to test for stationary failures.

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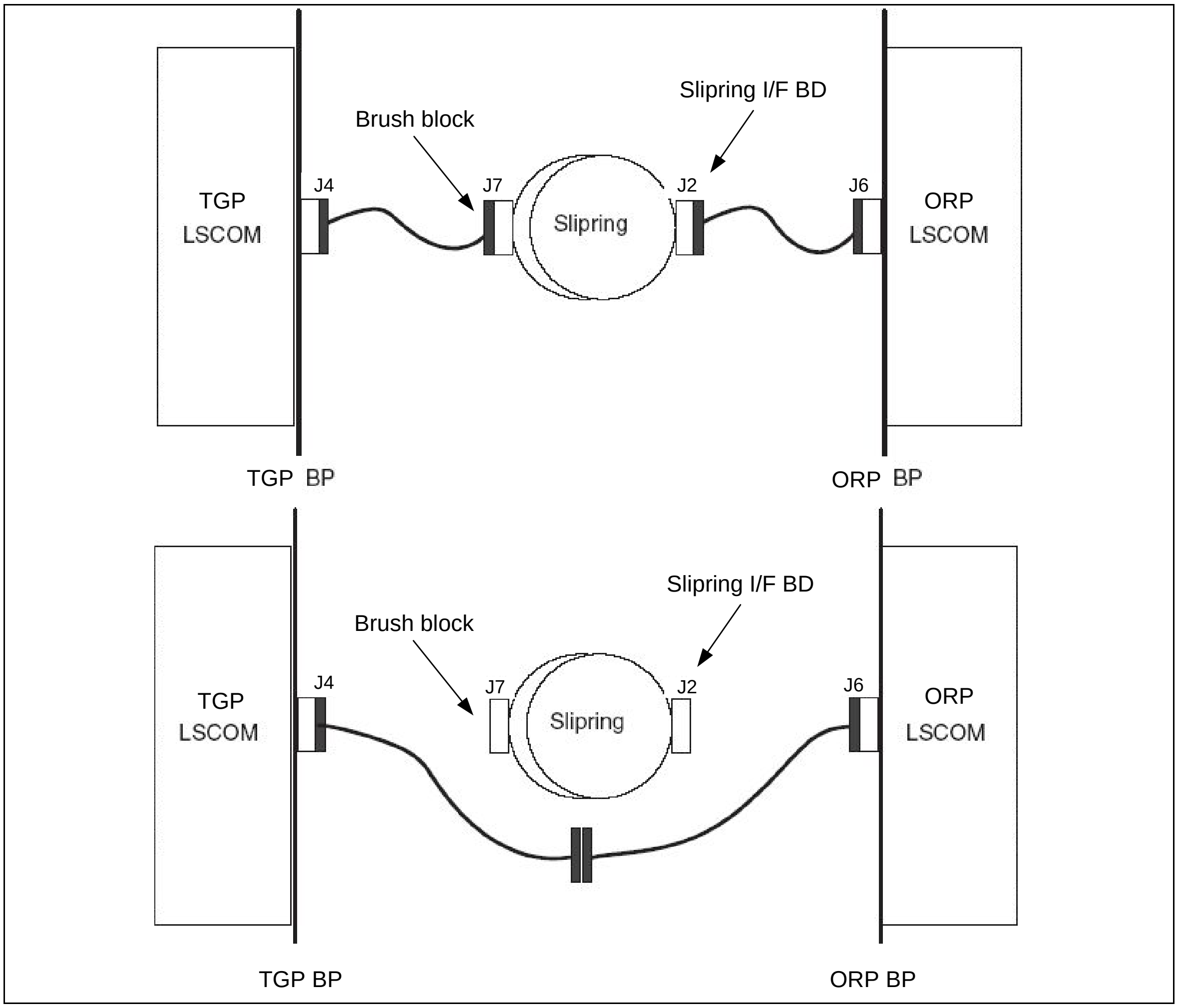
1-3 Slipring Bypass Procedure

Test for disconnects in a stationary condition by performing the bypass procedure shown in Illustration 3.

ILLUSTRATION 3 – BYPASS PROCEDURE



2. Remove Brush Block

1. Move table to its lowest elevation.
2. Remove gantry covers as required. Refer to the Service Methods manual: *Parts Replacement > Gantry > Enclosure > (Cover Removal Procedure).*

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NOTICE

Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

3. Turn **OFF** all 3 service switches (Axial Drive, HVDC, 120VAC) on the Service Switch Panel.

DANGER

ELECTROCUTION HAZARD
HIGH VOLTAGE PRESENT.
USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS. SEE SAFETY MENU OF SERVICE DOCUMENT GUI FOR APPROPRIATE PROCEDURES.

4. Perform LOTO for system power (A1 disconnect) prior to replacing this component.
5. Remove rear gantry cover support bracket.
6. Remove slipring covers.
7. Disconnect all connections to brush block (Illustration 4).

ILLUSTRATION 4 – SLIPRING BRUSH BLOCK ASSEMBLY

8. Remove four 6mm cap screws that secure brush block assembly to gantry.
9. Carefully remove brush block.

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3. Cleaning the Slipring

Follow this procedure to properly perform slipring cleaning.

NOTE: Use of Neoprene or nitrile gloves to limit irritation and ingestion of metallic dusts is recommended. Do not remove gloves near an exposed slipring. The powder inside the glove can contaminate the ring.

1. Clean off the dust from the slipring tracks and carefully around the block with a HEPA vacuum cleaner and the soft brush attachment.

   NOTE: The soft brush attachment that comes with the HEPA vacuum cleaner should NOT be used for anything other than slipring brush debris cleaning. The brush must be kept clean to avoid contamination of the slipring and brush tips. It is recommended to store the brushes or brush block in a separate closed bag (such as a plastic Zip-lock bag).

2. With the slipring eraser (Illustration 5), carefully rub the tracks using a large, sweeping motion.

ILLUSTRATION 5 – SLIPRING ERASER

IMPORTANT: Do not rub intensely in one spot for a long period of time.

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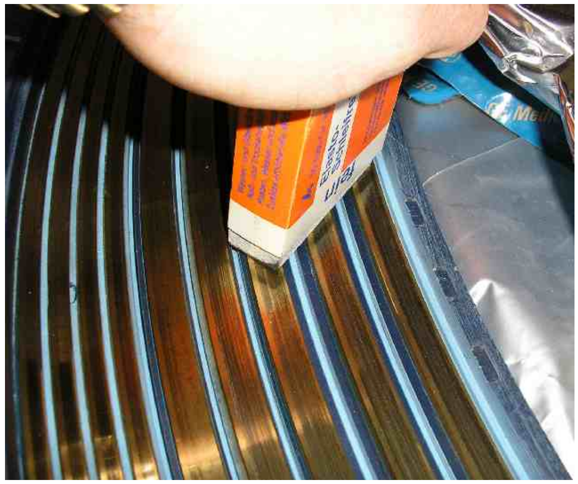
The ring color should return to a shiny new brass color (Illustration 6).

ILLUSTRATION 6 – CLEANING WITH THE ERASER



3. Clean the track again with the HEPA vacuum cleaner and the soft brush attachment.
4. Wipe the track with a soft white linen cloth while using the vacuum cleaner to remove any other debris.
5. Remove gloves and wash hands thoroughly after cleaning.

4. Replacing the Signal Brushes

NOTICE

Brush Contamination Possible. Do not touch brushes or rings with your fingers. Skin oils will contaminate the brushes and rings and reduce usable life and potentially create future failures.

1. Remove all individual signal brushes from the block by unscrewing brush cap and extracting brush.

   NOTE: Brushes are spring-loaded to ensure constant contact with the slipring during operation. When the block is removed, the springs relax, causing the brushes to bound outwards.
2. Install all new signal brushes on the block.

NOTICE

Brush Tip Damage Possible. Brush tips are extremely brittle. Do not apply sideways force as they will break. Any brush that has been damaged in this fashion must be replaced.

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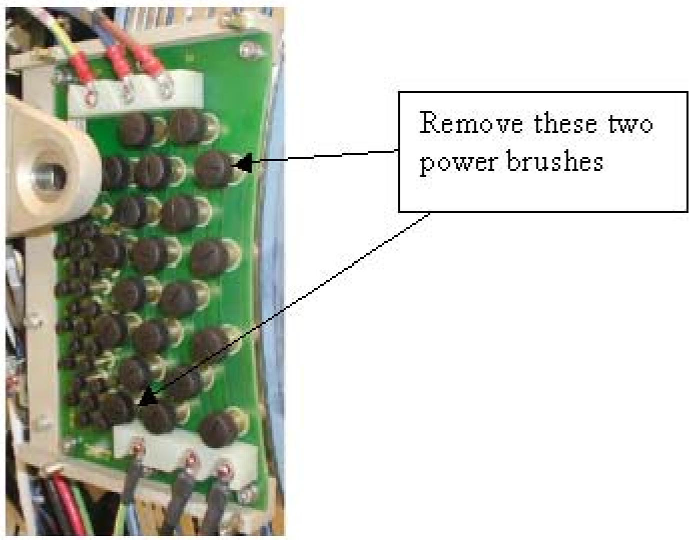
5. Aligning the Brush Block

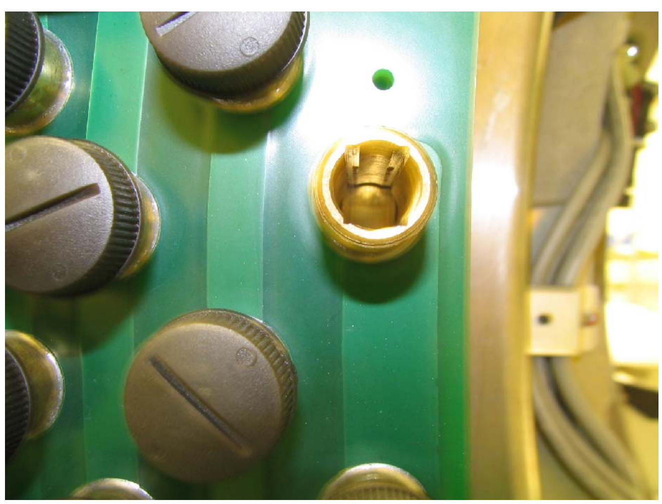
1. Carefully install brush block by exerting even pressure perpendicular to the ring surface.
2. Secure brush block with the four 6 mm cap screws. Do not tighten yet.
3. Carefully push brush block against the position adjustment set screws in the mounting bracket.
4. Remove two brushes from the inside HVDC ring top and bottom (Illustration 7). Remember their orientation for later installation.

ILLUSTRATION 7 – POWER BRUSHES TO USE FOR ALIGNMENT



5. Use a flashlight to verify that block position is adjusted so the brushes ride in the center of their tracks (Illustration 8).
   NOTE: The signal brushes transmit communications across the rings. As a final check, remove a few of them to ensure that they ride in the center of their tracks as well.

ILLUSTRATION 8 – CORRECT BRUSH BLOCK ALIGNMENT



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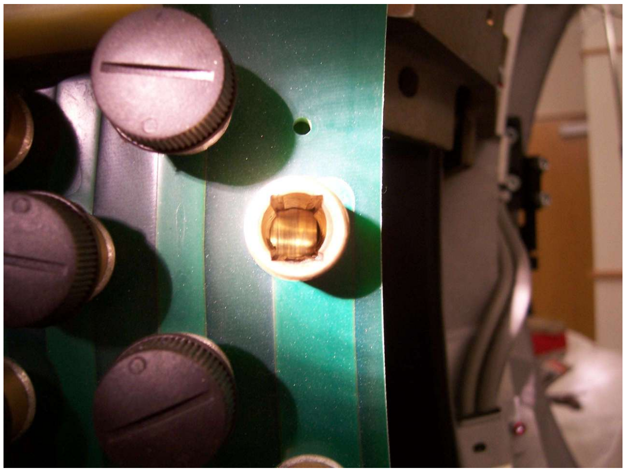
NOTE: Incorrect alignment occurs if either side of the track is visible (Illustration 9). If necessary, adjust the brush block setscrews to center the brush. To measure radial and axial runout, see the *Slipring Platter Replacement* procedure*,* Section 4.2, steps 15–18.

ILLUSTRATION 9 – INCORRECT BRUSH BLOCK ALIGNMENT



6. Torque four 6mm cap screws to 7.9 N-m.
7. Before reassembling the gantry, inspect the following:
   a. Check all power connections to the slipring – 208V on the back side of the ring.
   b. Check the brush block connections and visually inspection the harness with the sub-d connector.
   c. Check the small slipring i/f board on the rotating part of the ring.
   d. Verify that the 9-pin connector is securely connected to the board.
   e. Verify that all five 3mm cap screws are tight using a 2.5mm hex key.

   NOTE: Visually inspect all screws. In the past there have been occasions where screws were used that were too long.
8. Reassemble the gantry.

FINALIZATION

1. Document the number of gantry revolutions.
2. Hardware Reset using console gantry reset (Hardwire).
3. Acquire 10 scouts: (120kV/40mA, 1000mm table movement)
4. Acquire 100 axials: (120kV/80mA, 0.5 sec. Scan)
5. Acquire 1 helical: (120kV/40mA, 30 sec. Scan)
6. Acquire 10 axial scans: (120kV/400mA, 4 sec. Scan)
7. Verify NO increase in LSCOM errors.

Jason Katcha
Development Engineer – MICT-Eng-Hardware-Global Positioners

MI&CT Service Engineering Manager Signature____